Srijan Sharma

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.86
Kendriya Vidyalaya No.2 AFA Dundigal	Class XII (CBSE)	2024	93.80%
Kendriya Vidyalaya No.1 AFA Dundigal	Class X (CBSE)	2022	95.20%

SCHOLASTIC ACHIEVEMENTS

Awarded **the Academic Excellence Award** for securing the highest CG in the Biomedical Engineering batch at IIT Hyderabad, recognized on Foundation Day, April 2026

SKILLS

Programming Languages -- Python, C, C++

Tools & Frameworks -- PyTorch, Arduino, Figma, Wix Studio, LaTeX, After Effects

Domains --Machine Learning, RAG, LLM Evaluation, Prompt Engineering, UI/UX Design, 3D Modelling

Languages -- German (A2, Goethe-Zertifikat)

PROJECTS

ResearchLens -- Local LLM-Powered Research Assistant:

- **Built** a retrieval-augmented generation (RAG) system to close a research gap -- public literature APIs (PubMed, Europe PMC, OpenAlex, Semantic Scholar, bioRxiv) return abstract-only data -- by **engineering** in-browser PDF text extraction (PDF.js) for full-document analysis
- **Designed** a locally-hosted LLM pipeline (Ollama) generating structured JSON limitation/gap analysis, eliminating cloud dependency and API cost while keeping all document data client-side
- **Developing** an evaluation pipeline to quantify factual grounding of LLM-generated outputs against source text, addressing hallucination risk in generative research tools

GPT Language Model Implementation:

- **Implemented** a GPT-2-style transformer from scratch in PyTorch -- tokenization, positional embeddings, multi-head self-attention, layer normalization -- to **build** a first-principles understanding of the LLM architecture underlying ResearchLens
- **Verified** implementation correctness by validating output logit dimensions across batched inputs and testing layer normalization and masked self-attention against expected transformer behavior

Acoustic Spirometry for COPD Screening -- Independent Literature Review:

- **Conducted** an independent review of smartphone-based acoustic spirometry methods for low-resource COPD screening
- **Identified** critical validation gaps -- no COPD-majority cohort testing, >10% error rates, no unsupervised-use validation -- **informing** a proposed low-cost rural screening approach

OptiGrid:

- **Built** a linear-programming-based demand-scheduling optimizer to reduce industrial grid-carbon exposure, using real weather data (Open-Meteo API) and real Time-of-Day tariff structures (Telangana, TGERC FY2026-27)
- **Validated** ~7-12% CO2 reduction across 92 real historical days at both single-facility and 29-facility district scale, and identified/resolved a "rebound-peak" grid-overload risk created by synchronized responses to flat tariff incentives
- **Iteratively caught and corrected** multiple modeling errors (unrealistic baselines, double-counted solar credit, cost-blind scheduling) through rigorous validation against real data rather than accepting initial results at face value

RELEVANT COURSES

Biomedical Engineering -- Basic Bioinformatics, Medical Nanotechnology, Biomedical Imaging, Mathematical Modelling and Systems Biology, Biological Engineering

Programming & AI -- Introduction to Programming, Data Structures and Applications, Algorithms and Data Structures, Introduction to Modern AI, Artificial Intelligence

Mathematics -- Differential Equations, Transform Techniques, Elementary Linear Algebra, Introduction to Statistics, Introduction to Probability

Electronics -- Analog and Integrated Circuits, Introduction to Embedded Systems, Control Systems

POSITIONS OF RESPONSIBILITY

Core Member, Design Domain | Tinkerer's Lab, IIT Hyderabad

- **Designed** and prototyped the official Tinkerer's Lab website

Design Team Member | TEDx IIT Hyderabad

- **Designed** the intro sequence for the Spotlight Series (TEDx IIT Hyderabad)